

Document Scanner using OpenCV

Objective

Build a document scanner that automatically detects a document in an image, corrects its perspective, and generates a clean scanned version.

Image Preprocessing

Technique	Purpose
Grayscale Conversion	Simplify image processing
Noise Reduction	Remove unwanted noise
Contrast Enhancement	Improve visibility
Sharpening	Enhance document details

Edge Detection

Method: Canny Edge Detector

Purpose:

- Detect document boundaries
- Prepare image for contour detection

Result

The document edges become clearly visible, allowing accurate document extraction.

Document Detection & Scanning

Document Detection

Steps

1. Find contours
2. Sort contours by area
3. Approximate contours

4. Select the largest quadrilateral

Detection Criteria

Condition	Reason
4 Corners	Document shape
Convex Shape	Valid contour
Large Area	Ignore noise

Perspective Correction

Problem: Document is captured at an angle.

Solution:

- Detect corner points
- Apply Perspective Transform
- Generate top-down view

Result

Before	After
Skewed document	Flat document
Perspective distortion	Scanner-like output

Scanned Document Generation

Method: Adaptive Gaussian Thresholding

Benefits

- Removes shadows
- Handles uneven lighting
- Produces clean black-and-white output

Analysis & Results

Contour Analysis

Performed Operations

- Extract significant contours
- Calculate contour area
- Draw bounding boxes
- Ignore small contours

Shape Detection

Detected shapes when possible:

- Triangle
- Rectangle
- Square
- Pentagon
- Circle

Improvement

Additional filtering was applied to avoid false detections from text characters:

- Area filtering
- Aspect ratio filtering
- Solidity filtering
- Circularity filtering

Histogram Analysis

Observation

Two dominant peaks appeared:

Peak	Meaning
0	Text and dark regions
255	Document background

Conclusion

The clear separation between peaks indicates successful thresholding and strong document contrast.

Color Space Analysis

Color Space	Use
Grayscale	Edge detection and thresholding
HSV	Robust color representation

Image Comparison

Metrics Used

Metric	Purpose
MSE	Pixel difference
SSIM	Structural similarity

Result

- Low MSE indicates limited distortion.
- High SSIM indicates preserved document structure.